

F1 Race Strategy Intelligence Platform Report

The Problem?

- F1 is at the center of data and speed. Analysing huge loads of data in a matter of split seconds and win you the race or lose it. Pitstops, compound choice - everything plays an insanely huge role. So, completely understanding and analysing all the data available is the key.
- Analysis also depends on a ton of historical data and calculations as well. The data exists but it's buried in raw timing sheets that only engineers can read. Enthusiasts and hobbyists have no accessible way of exploring tire strategy decisions with real data.

The Users

- A dashboard for F1 enthusiasts and hobbyist analyst who want data backed answers about race strategy without writing the code.

What Success Looks Like?

- A casual user should be able to answer tire strategy questions using the dashboard without writing any code or having to read raw data.
- The degradation curves accurately reflecting the real lap time behavior across different compounds (validating against known race narratives)
- Driver consistency rankings will act like a 'sniff test' against known driver reputation and F1 knowledge.
- The AI chatbot should give answers grounded in actual race data and clearly state when it doesn't know something rather than providing inaccurate informations.

Questions the Tool will Answer

- How fast does each compound degrade and when does the performance 'cliff' hit?
- Which drivers were able to manage tire conditions consistently relative with the pace?
- How fast does each compound degrade?
- Did undercut/overcut payoff?
- How did the two drivers strategy compare and what was the time delta?
- When was the most optimal pit window vs when did the teams actually pit?

Phase 1

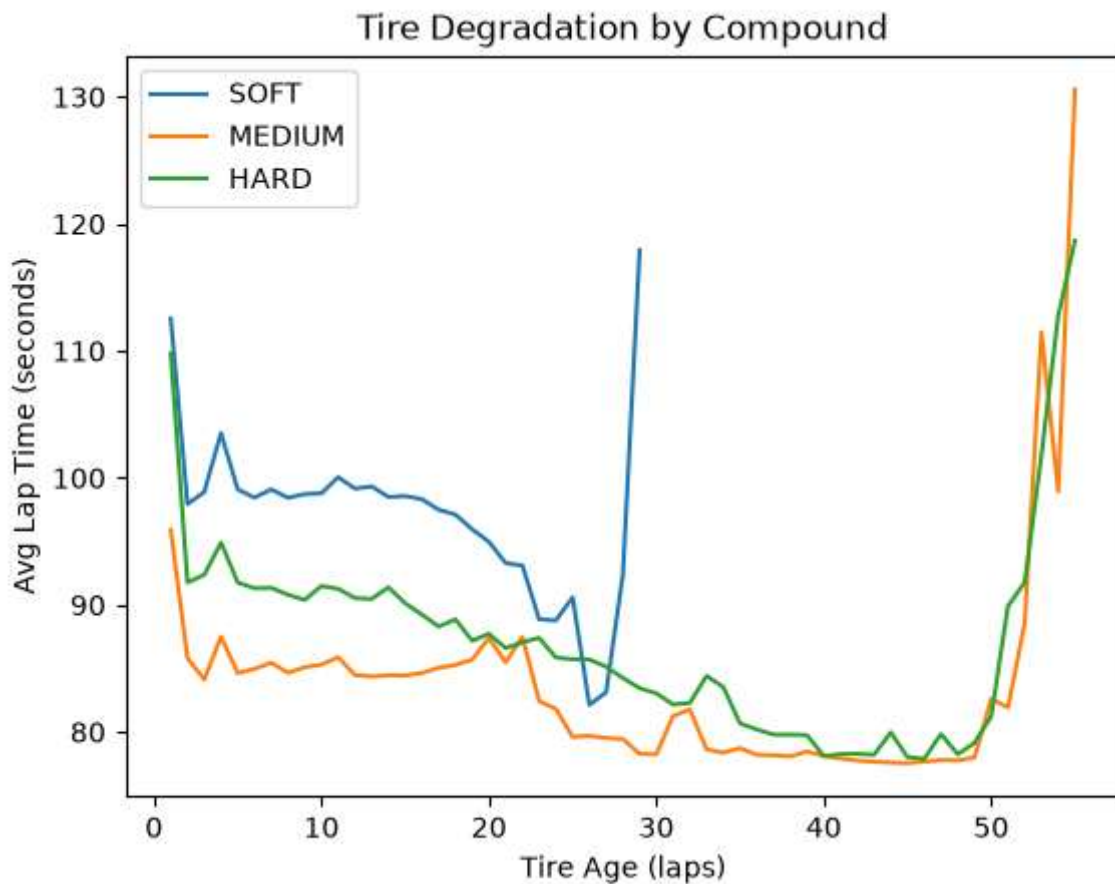


Image 1 - Tire Degradation by Compound: Shows how avg lap times goes up as tire age increases for soft, medium and hard tires